WundCheck

A mini CDSS that puts the daily wound check in the hands of the patient

We translated a guideline written for health professionals into an instrument for laypersons — and learned where that translation becomes dangerous.

48	**18**	**9**	**8**
data elements	decision rules	L1→L2 challenges	defects fixed

FHNW · MSc Medical Informatics · Clinical Decision Support Systems 2026

HEALTH SYSTEM NEED

Surgical site infections appear after discharge — when nobody is looking

60.1 %

of surgical site infections occur after discharge — systematic review of 1.4 million operations. The CDC surveillance window runs 30 days, 90 for implants.

Too late

Warning signs are read as normal. The infection progresses; treatment starts days later than it could.

Too early

Normal healing signs — a scab, day-2 redness — trigger calls that bind practice staff and worry patients.

SSI incidence in Europe 0.5–10.1 % depending on procedure (ECDC 2017: 10,149 SSIs in 648,512 operations) · 9.5 % after colon surgery, 1.0 % after hip replacement (WHO) · Switzerland: > 170 hospitals in the Swissnoso surveillance programme

Why a knowledge-based CDSS — and not the alternatives

Information leaflet	already exists everywhere; no feedback on the actual finding
Telephone follow-up	not deliverable daily; “the wound looks odd” is not usable data
Photo-based ML	light, angle and skin tone vary; decisions not explainable
Rule-based CDSS	traceable to a guideline, reviewable, maintainable by clinicians

Scope

Adults · elective surgery · wounds with primary closure · any body region.

Not covered

Chronic wounds · burns · secondary healing · children · contaminated emergencies

Woelber et al., Surg Infect 2016;17(5):510–19 · ECDC Annual Epidemiological Report 2017 · WHO Global Guidelines for the Prevention of SSI · CDC/NHSN SSI Event

Three users on three levels — only one of them operates the app

1 **Peter Brunner, 58**
Patient · primary user

Knee replacement 4 days ago. Little medical knowledge, reading glasses, uses everyday apps.

"I want to know it is fine — without calling anyone."

2 **Sandra Meier, 34**
Practice assistant · secondary

Triages calls and messages all day. High workload, no time for guessing games.

"I need to see in 10 seconds whether this needs an appointment today."

3 **Dr Anna Keller, 45**
Surgeon · tertiary

Operates and runs follow-ups. Wants the trajectory, not a retrospective recollection.

"I want the full trend without opening every single answer."

SCENARIO A **Day 6 — the scab**

36.9 °C, pain 2/10 and better than yesterday — but a brown scab has formed, and Peter's brother once had a wound infection. WundCheck returns GREEN with "a scab is a normal sign of healing — do not pick it off". Peter does not call the clinic. The entry still goes into the record, so Dr Keller sees an unbroken trajectory at the suture removal appointment.

The call that is avoided is worth as much as the one that is triggered.

Four sources, three roles — and one we deliberately discarded

CDC / NHSN	Normative — definition	Superficial vs. deep incisional SSI, 30/90-day window, the five clinical signs	*Surveillance definition*
NICE NG125	Normative — management	Post-operative wound care; §1.1.3 requires patients be taught to recognise an SSI	*Evidence-based guideline*
Southampton score	Instrument	Visual grading 0–V — the anchor for our image selection	*Validated instrument*
Campwala et al. 2019	Evidence	Compares CDC, ASEPSIS and Southampton	*Retrospective, n = 22*

Why we discarded ASEPSIS — the best performer

It needs antibiotic administration, drainage, debridement, inpatient days and culture results. None of that is obtainable by a patient at home. We chose the usable instrument over the statistically better one — and we say so.

How good is our evidence? Not very — and that shapes the use

Campwala et al.: n = 22, retrospective, single-centre, implant-based breast reconstruction — a population far from ours, and the endpoint is implant failure, not detection of infection. Southampton showed only limited predictive value there.

So we use it as a communication and triage grid, not as a prognostic instrument.

Our image selection is a published grading, not a UI whim

0	Normal healing	→ Calm & dry
I	Mild bruising and/or erythema	→ Slightly reddened
II	Erythema plus other signs of inflammation	→ Markedly reddened & swollen
III	Clear / serous / bloody discharge	→ Clear or bloody discharge
IV	Pus	→ Cloudy / purulent discharge
V	Deep infection ± tissue breakdown	→ Open / pus

What this buys us

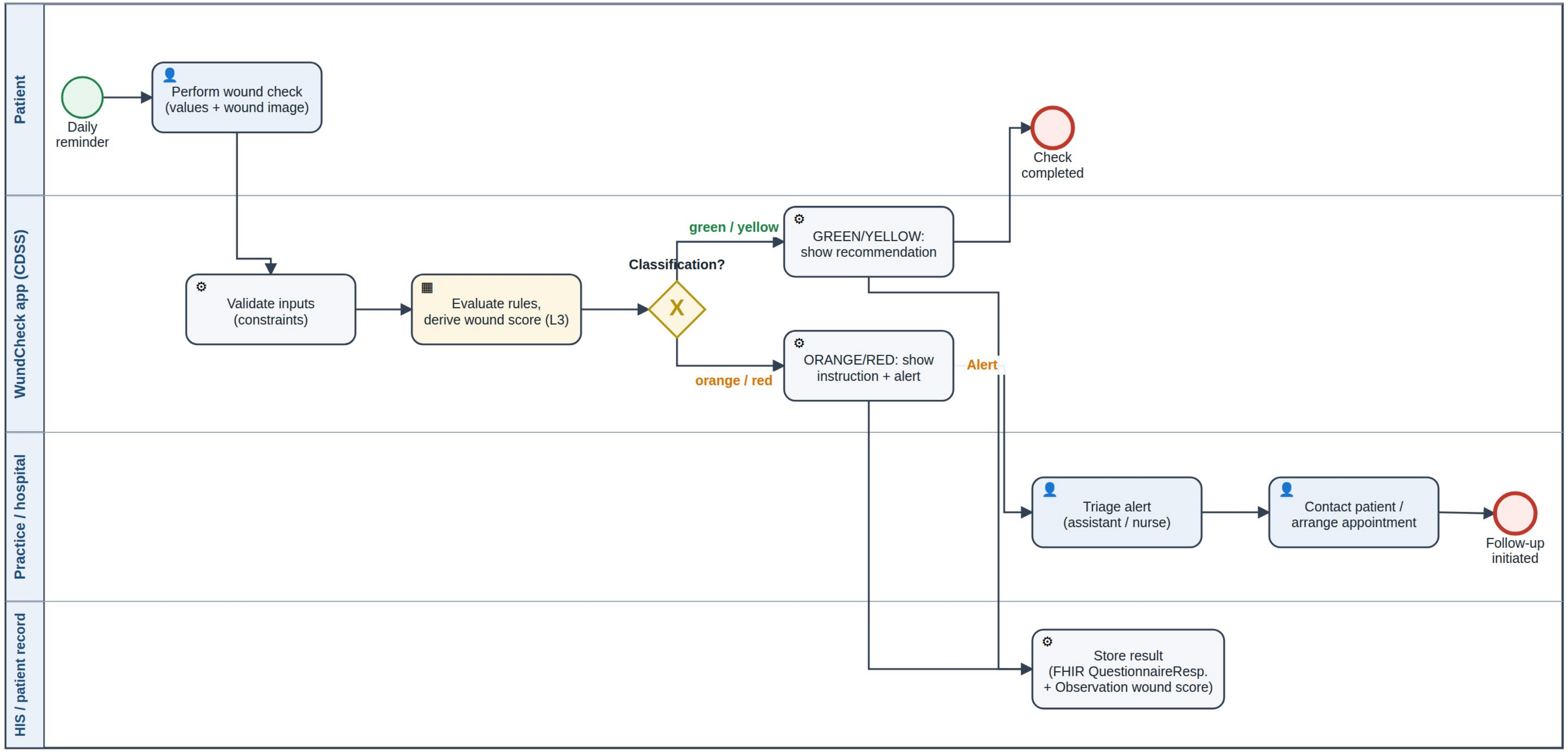
- a citable source for our core interaction
- a non-trivial derived element
- an honest, evidenced limitation

The adaptation problem

Southampton grades "> 2 cm along the wound". Patients do not measure. We ask "one small spot or the whole wound?" — closer to usable, further from precise.

The logic sits in a business rule task — not in the app

WundCheck — High-level workflow (BPMN): daily wound check after surgery with primary closure



Legend: 👤 User Task · ⚙ Service Task (automated) · ▦ Business Rule Task (decision table L3) · X = exclusive gateway — editable source: workflow-highlevel.bpmn (bpmn.io / Camunda Modeler)

48 data elements — every one traceable to a guideline

48 data elements	**9** derived	**38** terminology-bound	**47** FHIR-mapped

ID	ELEMENT	TYPE	TERMINOLOGY	FHIR	L1 SOURCE
DE-020	Body temperature	decimal	LOINC 8310-5	Observation.valueQuantity	CDC — fever > 38 °C
DE-030	Wound appearance	select_one	SNOMED 225552003	Observation.valueCodeable	Southampton 0–V
DE-032	Red streak	select_one	SNOMED 46117009	Observation.valueCodeable	NICE NG125 §1.4.9
DE-035	Discharge	select_one	SNOMED 449736006	Observation.valueCodeable	CDC — purulent drainage
DE-045	Southampton grade	calculated	SNOMED 225552003	Observation.valueCodeable	Southampton

Sequence follows the patient, not the guideline: baseline data once, then “how do I feel”, then the wound — because inspecting it means opening the dressing.

Worst-first: a reassuring finding can never overwrite a warning

RED · emergency
R1 R2 R3
Systemic signs · lymphangitis · dehiscence or pus

▼

ORANGE · suspected
R9 R4 R5 R11 R12
Feeling unwell · purulent discharge · fever · persistence

▼

YELLOW · abnormal
R6b R6a R7 R10
Inflammation · wound gap · bleeding on anticoagulants

▼

GREEN · all clear
R8
Catch-all — never “everything is fine” without a contact hint

Why worst-first

A false alarm costs a phone call. False reassurance can delay an infection by days. We accept a higher false-alarm rate on purpose.

Two-pass evaluation

One rule needs the classification as its own input. The rules run twice — the worse result wins, so a second pass can escalate but never de-escalate.

All thresholds are named constants in the L3 with their source — a guideline change is one line, not a release.

Nine places where the guideline left us on our own

C-01 Professional instrument → layperson instrument	C-02 Sub-grades without centimetres	C-03 When does redness stop being normal?
C-04 Scab: warning or reassurance?	C-05 CDC criteria assume a clinical examination	C-06 Southampton as triage, not prognosis
C-07 Two fever thresholds instead of one	C-08 No trajectory on the first day	C-09 Severity scale ≠ urgency of action

The two highest-risk decisions sit in the same place

C-01 Southampton was built for trained staff. We turned it into an image selection for laypersons. That is our core idea — and it is unvalidated.

C-03 No source defines the day on which redness stops being physiological. We set day 3. Too early means false alarms; too late means delayed detection.

All nine would need clinical sign-off. We do not have it — and we list that as a limitation.

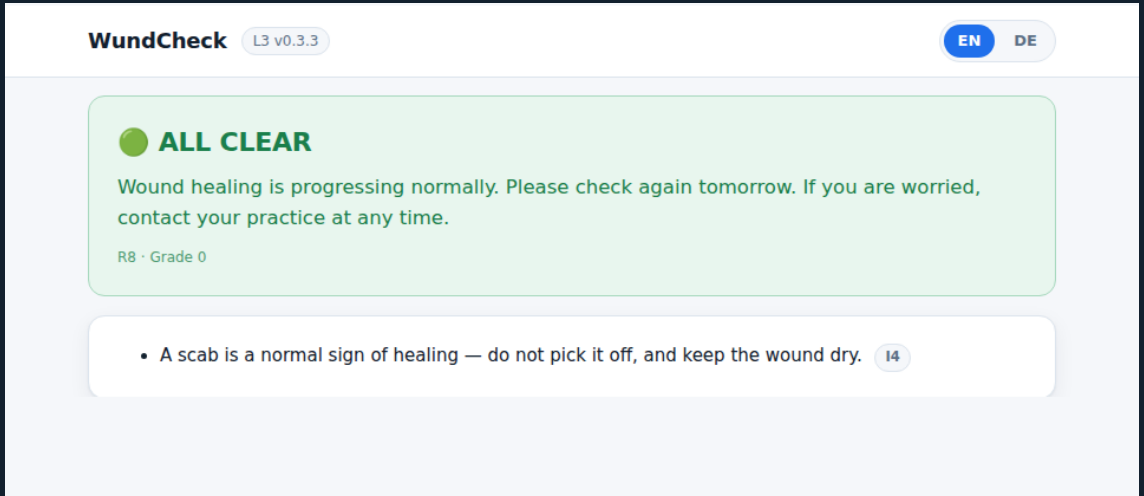
Author in readable JSON, speak standards at the interface

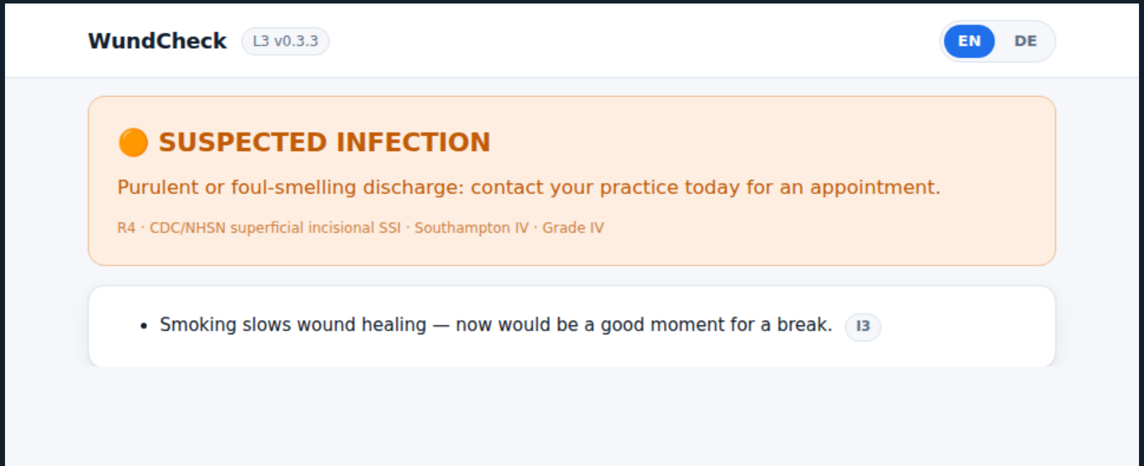
XLSForm / ODK	Established, executable by existing engines	Standard widgets cannot carry our image cards — the L4 would dictate the UX	
FHIR Questionnaire + CQL	Full standard, WHO SMART compliant	“FHIR resources are not human readable” — not reviewable by our clinical lead in six days	
Hardcoding in the app	Fastest to a prototype	Every rule change needs a developer and a rebuild	
Custom JSON + FHIR at the edge	Reads like the L2; full control of the UX; standards where they count	No existing engine runs it directly; expression language is not CQL	✓

And we can prove it: a check scans the app source for every element id, threshold, code system and endpoint. Any clinical content in the L4 fails the build.

DEMO

One patient, two days — and a threshold change with no rebuild





Demo video: submission/WundCheck_L4_Demo.mp4 · 1:46 · Single HTML file · offline · no installation · EN / DE

1

Day 6 — GREEN · Scenario A

A scab worries the patient. The system reassures and names the reason. No call to the clinic.

2

Day 7 — ORANGE · Scenario B

Purulent discharge, 38.3 °C. R4 fires, the patient is told to contact the practice today, a FHIR Communication goes out.

3

Threshold change

fever_orange 38.0 → 37.5 in the L3 inspector. Same case, different result. No rebuild, no developer, not a line of code.

Why the interface looks like this

- Images, not clinical terms — crosses language and literacy barriers
- The result names its reason, not just a colour — against automation bias
- Reassurance is a first-class output, not a leftover
- Baseline data once only — a daily check has to stay under 3 minutes

QUALITY ASSURANCE

Two defects were invisible until the rules actually ran

🔴	**B-01**	Marked inflammation on day 1–2 classified GREEN	L2
🔴	**B-02**	“I feel really unwell” triggered nothing at all	L2
🔴	**B-03**	Emergency rule unreachable behind conditional logic	L2/L3
🟤	**B-04**	Fever threshold duplicated in two places	L3
🟢	**B-05**	Anticoagulant status collected but never used	L2
🟤	**B-06**	Southampton IV mapped RED in one doc, ORANGE in another	L2
🔴	**B-07**	Trend rule could never fire — circular dependency	L3/L4
🟤	**B-08**	That same rule was unreachable even after the fix	L2

19/19 decision-logic tests

20/20 FHIR conformance checks

6 hazards ranked

Residual risk

We never tested the one thing our whole approach rests on: whether laypeople assign the images correctly. That is monitored after deployment — GREEN cases that still present to a physician within 72 h.

Risk-based testing per Day 5: identify hazards → rank by risk → test critical paths first → document residual risk.

How it reaches users, how it is maintained, how we would know it works

0 · Preparation	1 · Pilot	2 · Evaluation	3 · Expansion	4 · Scaling
clinical sign-off	1 hospital, 30 patients	rule revision, UX rework	HIS integration live	further sites, languages
4 wk	*3 mo*	*2 mo*	*6 mo*	*open*

Implementation

Provision
QR code on the discharge sheet — no app store, no account, no install

Training
Patient 5 min at discharge (first check done together) · practice 30 min · onboarding module for new staff

Maintenance
Clinical department owns the content; four-eyes sign-off; guideline update to live in ≤ 4 weeks

Integration
FHIR R4 to the HIS / Swiss EPD; endpoint is L3 configuration

Monitoring

Functionality
> 99 % error-free runs

Stability
zero data loss on network drop

Fidelity
> 70 % of days checked

Data quality
< 5 % incomplete entries

Evaluation

Pilot
SUS + patient-vs-clinician agreement (κ)

Demonstration
chart review — time to medical contact

Scale-up
stepped-wedge; SSI rate, readmissions

Adverse effects
false alarms, alert overrides, equity

Interoperability reached: structural yes · semantic partly · organisational no. · Most important safety indicator: GREEN cases that still present to a physician within 72 h.

False reassurance weighs more than a false alarm

Benefits

- Earlier treatment when an infection develops
- Fewer contacts without clinical consequence
- Complete trajectory for the follow-up appointment
- Structured, standardised follow-up data

Risks

- Automation bias — the patient stops observing
- Alert fatigue in the practice
- Scaling our own errors consistently, at volume
- Digital divide — the gap may widen

Is WundCheck a medical device? Yes — SaMD, IMDRF Category II

State of the healthcare situation = serious: an untreated SSI is not self-limiting and can lead to sepsis or revision surgery. Significance of information = drive clinical management: we neither diagnose nor treat, but we determine whether and when clinical action is taken.

Standards that would apply: ISO 14971 · IEC 62304 · IEC 62366 · ISO 13485. The EU AI Act does not — the system is rule-based, a deliberate consequence of avoiding image analysis.

	Inform	Drive	Treat/dx
Critical	IV	III	II
Serious	III	**II**	I
Non-serious	II	I	I

IMDRF risk categorisation

Individual contributions

Clinical lead Name ______________	L1 sources & evidence appraisal · health need · personas · L1→L2 challenges	Slides 2–5, 9
L2 lead Name ______________	BPMN · data dictionary · decision table · terminology mapping	Slides 6–8
Tech lead Name ______________	L3 schema · application · FHIR interface · demo	Slides 10–11
QA & implementation lead Name ______________	hazard analysis · test protocol · bug log · M&E · impact · regulation	Slides 12–14

Every member can answer questions on the whole project, not only their own part.

Decision table — all 18 rules

R1	temp ≥ 39 OR chills = 1	CDC · sepsis red flags
R2	red_streak = 1 OR redness_extent = 4	NICE NG125 §1.4.9
R3	wound_open = 2 OR bleeding = 2 OR wound_state = 4	CDC deep incisional
R9	wellbeing = 2 OR nausea_vomiting = 2	systemic signs
R4	secretion = 3 OR secretion_smell = 1	CDC superficial incisional
R5	fever OR (pain ≥ 7 AND worsening AND day ≥ 3)	CDC · NICE
R11	secretion ≥ 1 AND discharge ≥ 3 days	Southampton IIId
R12	classification = yellow AND yesterday = yellow	persistence
R6b	Southampton II OR swelling = 3 (from day 1)	Southampton II
R6a	Southampton I AND day ≥ 3	wound healing phases
R7	wound_open = 1 OR numbness OR mobility change	follow-up standards
R10	bleeding ≥ 1 AND anticoagulants	bleeding assessment
R8	catch-all	—
I1–I5	risk factors · medication · smoking · scab · high-risk procedure	non-exclusive

Test protocol — 19 decision-logic cases, all passing

T1	R1 fever · temp 39.2	RED (R1)	✓
T2	R1 boundary · temp 39.0	RED — inclusive	✓
T3	R3 safety · image “open/pus” alone	RED (R3)	✓
T4	R5 before R6	ORANGE (R5)	✓
T5	mild signs on day 2	GREEN	✓
T6	conditional logic	3 follow-ups hidden	✓
T7	constraint · temp 45	input rejected	✓
T8	info rules non-exclusive	GREEN + I1 + I2 + I3	✓
T9	scab reassures	GREEN + I4	✓
T10	worst-first vs reassurance	ORANGE (R4) + I4	✓
T11	B-02 fixed · feeling unwell	ORANGE (R9)	✓
T12	B-03 fixed · red streak	RED (R2)	✓
T13	B-01 fixed · grade II on day 2	YELLOW (R6b)	✓
T14	B-01 counter-check	GREEN	✓
T15	Southampton derivation	grade IV → ORANGE	✓
T16	B-05 fixed · bleeding + anticoag	YELLOW (R10)	✓
T17	persistence rule	ORANGE (R12)	✓
T18	discharge ≥ 3 days	ORANGE (R11)	✓
T19	threshold change without rebuild	GREEN → ORANGE	✓

Reproduce: node tools/run-tests.mjs · node tools/check-fhir.mjs · node tools/check-hardcoding.mjs

Hazard analysis and risk matrix

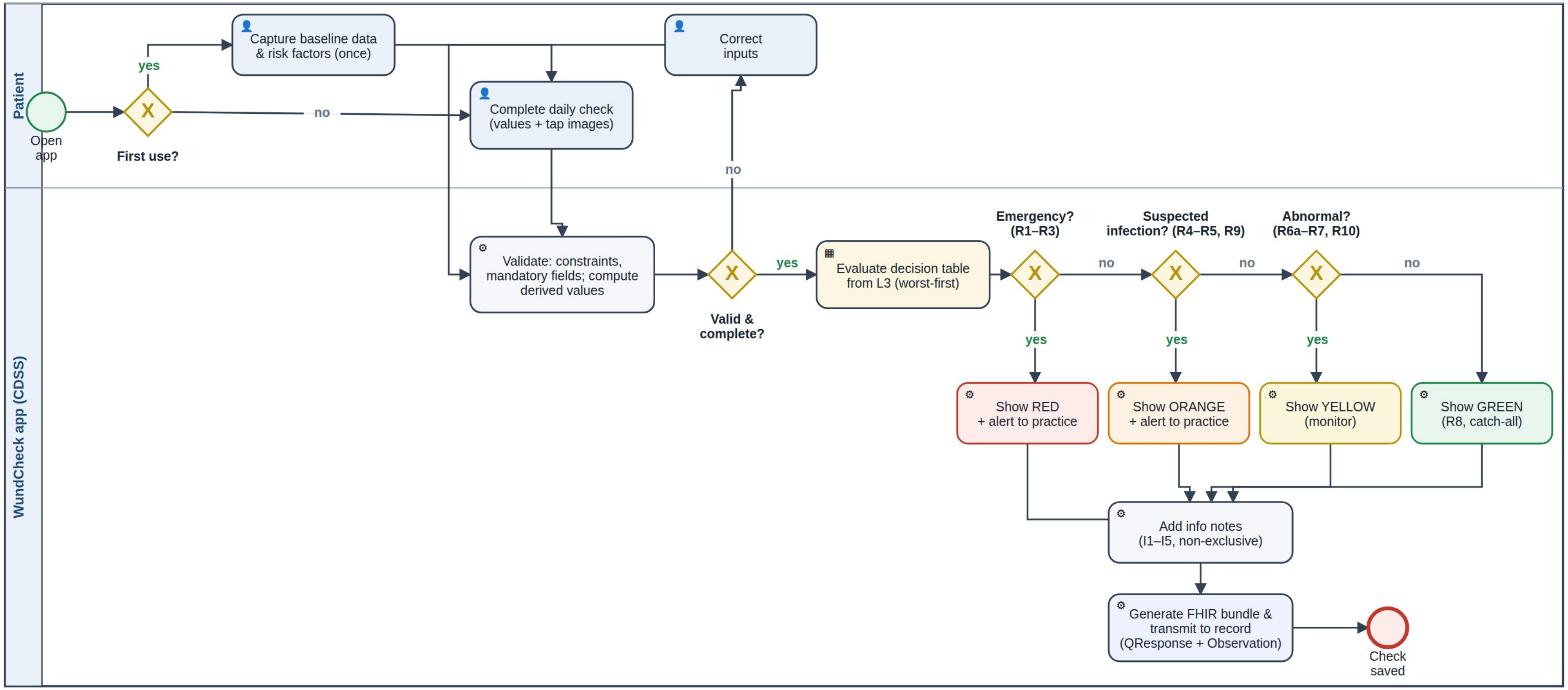
H-01	An infection is classified GREEN	sev 5 · lik 3	15
H-02	An emergency is classified ORANGE	sev 5 · lik 2	10
H-03	The patient abandons the check	sev 3 · lik 3	9
H-04	False alarm during normal healing	sev 2 · lik 4	8
H-06	The practice overlooks an alert	sev 4 · lik 2	8
H-05	An implausible input is processed	sev 3 · lik 2	6

Severity \ Likelihood	1	2	3	4	5
5	5	**H-02**	**H-01**	20	25
4	4	**H-06**	12	16	20
3	3	**H-05**	**H-03**	12	15
2	2	4	6	**H-04**	10
1	1	2	3	4	5

Score ≥ 10 → full test coverage. Score 6–9 → sample-based coverage.

Detailed process inside the app

WundCheck app — detailed process of one daily check (BPMN): validation, worst-first decision logic, FHIR transmission



Gateway cascade = decision logic in worst-first order (safety principle): emergency before suspected infection before abnormal; GREEN only as catch-all. Editable source: workflow-app-detail.bpmn

BACKUP · L3 / L4

The FHIR interface — a real transaction bundle, not a download

QuestionnaireResponse	every answer, typed and terminology-coded; canonical carries the L3 version	*every run*
Observation	traffic light, Southampton grade, triggering rule, L3 version, days post-op	*every run*
Communication	structured alert, priority urgent (ORANGE) or stat (RED)	*ORANGE / RED only*

The endpoint is configuration, not code

```
"fhir": { "endpoint": "https://…/baseR4", "patient_reference": "Patient/example",
          "alert_recipient": "Organization/practice", "send_mode": "transaction" }
```

Redirecting to a hospital FHIR server or the Swiss EPD gateway is one line of JSON. If the endpoint is unreachable the bundle queues — the patient still gets the recommendation.

BACKUP · METHOD

Traceability — one finding, all the way through

L1	CDC/NHSN — “purulent drainage from the superficial incision”
▼	
L2	DE-035 discharge · SNOMED 449736006 · rule R4 → ORANGE
▼	
L3	"when": "secretion = 3 or secretion_smell = 1"
▼	
L4	image selection “cloudy / purulent” → SUSPECTED INFECTION
▼	
QA	T10, T15 — passed · challenge C-09 documents the severity/urgency split
▼	
FHIR	QuestionnaireResponse.item[secretion] + Observation.component

Every element, rule, challenge, hazard, test and defect carries an ID that is identical in all artefacts.

The L3 in one screen — this file is the CDSS

```
{ "meta": {
    "version": "0.3.3",  "languages": ["en","de"],  "default_language": "en",
    "l1_sources": [ CDC/NHSN · NICE NG125 · Southampton · Campwala 2019 ],
    "thresholds": { "fever_orange": 38.0,   // CDC – deep incisional SSI
                    "fever_red":    39.0,   // group decision – challenge C-07
                    "normal_healing_days": 3 },  // group decision – C-03
    "fhir": { "endpoint": "…/baseR4", "send_mode": "transaction" } },

  "elements": [ {
      "id": "secretion",  "de_id": "DE-035",  "type": "select_one",
      "label":   { "en": "Is fluid coming out of the wound?",
                   "de": "Kommt Flüssigkeit aus der Wunde?" },
      "options": [ 0 = none · 1 = clear · 2 = bloody · 3 = purulent, each with an image ],
      "terminology": { "snomed": "449736006" },
      "fhir": "Observation.valueCodeableConcept",
      "l1_source": "CDC/NHSN – purulent drainage · Southampton III–IV" } ],

  "decision_rules": [ {
      "id": "R4",  "priority": "orange",  "exclusive": true,
      "when":   "secretion = 3 or secretion_smell = 1",
      "output": { "en": "SUSPECTED INFECTION – contact your practice today…" },
      "annotation": "Purulent drainage is a primary criterion of superficial SSI.",
      "reference":  "CDC/NHSN superficial incisional SSI · Southampton IV" } ] }
```

Bilingual by construction

Every user-facing string is a language-keyed object. A further language is a content change, not a code change.

Thresholds carry their source

Each constant names where it comes from — a guideline value or a documented group decision.

Rules carry their reference

Every rule states its L1 source and an annotation, so a clinician can review it without reading code.

The endpoint is configuration

Redirecting to a hospital FHIR server is one line — no code change, no rebuild.

48 elements · 18 rules · 54 KB · read by the L4 at runtime — swap this file and the app becomes a different CDSS.

Data dictionary — the four required columns

ID	LABEL	TYPE	RESPONSE OPTIONS	CONDITIONAL LOGIC	CONSTRAINT	TERMINOLOGY
DE-020	Body temperature today	decimal	free numeric input, °C	—	34–43 °C	LOINC 8310-5
DE-023	Pain 0–10	integer	NRS 0–10	—	0–10	LOINC 72514-3
DE-026	How do you feel overall?	select_one	0 = well · 1 = tired · 2 = really unwell	—	—	SNOMED 386661006
DE-030	Which picture matches your wound?	select_one	1 = calm & dry · 2 = slightly red · 3 = markedly red & swollen · 4 = open / pus	—	—	SNOMED 225552003
DE-032	Does a red streak run away?	select_one	1 = yes · 0 = no	none — deliberately ungated (B-03)	—	SNOMED 46117009
DE-035	Is fluid coming out?	select_one	0 = none · 1 = clear · 2 = bloody · 3 = cloudy / purulent	—	—	SNOMED 449736006
DE-036	Does the discharge smell?	select_one	1 = yes · 0 = no	secretion ≥ 1	—	SNOMED 288227007
DE-037	One spot or the whole wound?	select_one	1 = small spot · 2 = along the wound	secretion ≥ 1	—	—
DE-038	For how many days?	integer	free numeric input, days	secretion ≥ 1	0–60	—
DE-025	Did pain disturb your sleep?	select_one	1 = yes · 0 = no	pain ≥ 4	—	SNOMED 301345002
DE-045	Southampton grade	calculated	0 · I · II · III · IV · V	derived from 5 elements	—	SNOMED 225552003

48 elements in total · 9 derived · 4 with conditional logic · 8 with constraints · 38 with a terminology binding · 47 with a FHIR mapping. Full table: docs/05.

BACKUP · DOCUMENTATION

Depth that did not fit into ten minutes

Testing types applied (Day 5 vocabulary)

- L2 verification — decision table ↔ BPMN cascade ↔ L3 rules reconciled
- Functional testing — individual rules, exclusions, severity order, boundary values
- Integration testing — derived elements → rules → output; conditional logic → rule reachability
- Usability testing — image assignment by lay people: NOT carried out, our largest residual risk

The L3 implementation package (Day 3)

- Documentation — health need, personas, BPMN
- Content / workflow — sections, relevant, decision rules
- Terminology — SNOMED CT / LOINC / ICD-10 bindings
- Validation — 19 test cases, test data, conformance checks
- Interfaces — FHIR export · UI/UX — image options, traffic light

Stakeholders and where it could fail

- The critical group is the practice, not the patient
- If Sandra experiences the alerts as a burden, the rollout fails — however good the classification is
- Hence: alerts only from ORANGE upwards, structured instead of free text, alert load measured explicitly

What we would do with three more months

- 1 · Clinical sign-off on every threshold and rule — it settles all nine L1→L2 decisions at once
- 2 · Usability study on the image assignment — validates the central design idea
- 3 · Graphics for several skin tones and body regions
- 4 · Convert the L3 to a FHIR Questionnaire — the software neutrality we consciously gave up

Sources

Woelber E et al.	Proportion of Surgical Site Infections Occurring after Hospital Discharge: A Systematic Review. Surg Infect 2016;17(5):510–19.	*60.1 % post-discharge*
ECDC	Healthcare-associated infections: surgical site infections — Annual Epidemiological Report 2017.	*0.5–10.1 % by procedure*
CDC / NHSN	Surgical Site Infection Event (SSI). Patient Safety Component Manual.	*30 / 90-day window*
NICE	Surgical site infections: prevention and treatment. NG125, 11 Apr 2019, upd. 19 Aug 2020.	*nice.org.uk/guidance/ng125*
Bailey IS et al.	Southampton Wound Scoring System.	*grades 0–V*
Campwala I, Unsell K, Gupta S	A Comparative Analysis of Surgical Wound Infection Methods: CDC, ASEPSIS and Southampton in breast reconstruction SSI. Plastic Surgery, 2019.	*PMID 31106164*
WHO	Monitoring and Evaluating Digital Health Interventions: a practical guide. 2016.	*maturity stages*
WHO	SMART Guidelines — the L1–L5 knowledge layers.	*who.int*
IMDRF	Software as a Medical Device: possible framework for risk categorisation.	*categories I–IV*
HL7	FHIR R4 — QuestionnaireResponse, Observation, Communication.	*hl7.org/fhir/R4*